FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Vacuum Extraction (Ventouse)

Clinical skill | Week 4 · Hypertension, twins & the pelvic floor | Prof Mirembe

Vacuum extraction (ventouse) is an assisted vaginal birth in which a suction cup placed on the fetal head at the flexion point lets the operator add traction to the mother's pushing. It is safe only when every prerequisite is met (fully dilated, membranes ruptured, head ≤1/5 palpable and at or below the spines, position known), when traction follows the pelvic curve with contractions, and when the operator stops early if there is no descent.

Must know before the ward round

1. Assisted vaginal delivery is one of the 7 BEmONC signal functions, so it should be available at every HC III and above. In Uganda it is badly under-used, which pushes women into risky second-stage caesareans.
2. Prerequisites. The cervix is FULLY dilated and the membranes ruptured. The head is ≤1/5 palpable abdominally, with the bony skull at or below the ischial spines. The position is known precisely. Moulding and caput are assessed, the bladder is empty, and there is consent, analgesia, a trained operator, a newborn resuscitator and a back-up plan (CS).
3. The flexion point is on the sagittal suture about 3 cm in front of the posterior fontanelle (about 6 cm behind the anterior fontanelle). Centre the cup there for a flexing median application, so the smallest diameter (suboccipitobregmatic, 9.5 cm) leads.
4. Working pressure is about 0.8 kg/cm² (about 600 mmHg, 80 kPa). Pull only with contractions and maternal effort, along the pelvic axis (J-shaped curve), perpendicular to the cup, with the non-pulling hand checking the cup.
5. Abandon if there is no descent with any correctly applied pull, delivery is not imminent after 3 pulls, the cup detaches twice, or about 15–20 minutes have passed. Avoid sequential instruments where possible.
6. Contraindications include gestation below 32 weeks (caution at 32–36), face/brow/breech presentation, a head that is not engaged or ≥2/5 palpable, an incompletely dilated cervix, an unknown position, suspected CPD (3+ moulding), and fetal bleeding or bone disorders.
7. Compared with forceps, vacuum fails more often and causes more cephalhaematoma and retinal haemorrhage, but causes less maternal perineal trauma (third/fourth-degree tears) and needs less analgesia.
8. Subgaleal haemorrhage (bleeding under the scalp aponeurosis, which crosses suture lines) can hide a large part of the baby's blood volume and kill within hours. Watch every vacuum baby for a boggy, spreading scalp swelling, pallor, tachycardia and increasing head circumference.
9. Examine the abdomen before the VE. If fifths palpable and VE station disagree (e.g. 2/5 palpable but 'station +2'), trust the abdomen, because caput misleads the VE.
10. RCOG levels. Outlet (scalp visible without parting the labia) and low (bony skull at +2 or below, 0/5 palpable) can be done in the delivery room. Mid-cavity (1/5 palpable, station 0 to +1) is a trial in theatre by an experienced operator. High (2/5 or more palpable) means CS.
11. In the second stage check the FHR every 5 minutes (after each contraction; normal 110–160 bpm), and between pulls. A deteriorating FHR with birth not imminent means stop and go to CS.
12. After every ventouse, give oxytocin 10 IU IM, do a PR exam for OASI, make sure she voids within about 6 hours, and give the baby vitamin K 1 mg IM with scalp checks and head circumference at least hourly for the first few hours.

1 Purpose and Indications
2 Contraindications and Precautions
3 Relevant Anatomy and Normal Values
4 Equipment and Preparation
5 Procedure Step by Step
6 Findings and Interpretation
7 Clinical Acumen
8 Complications and Troubleshooting
9 Documentation and Presenting Findings
10 OSCE Checklist
11 Key Points
12 References

1. Purpose and Indications

Vacuum extraction (ventouse) means applying a suction cup to the fetal scalp and pulling, in time with contractions and maternal pushing, to expedite a vaginal birth in the **second stage** of labour. It helps a baby that is **almost there**. It does not pull a baby through a pelvis it cannot fit.

In Uganda, ventouse is a **life-saving but neglected skill**. A woman in prolonged second stage at an HC III or HC IV with a low head and a distressed baby can often be delivered in **minutes** with a ventouse. The alternative is a long referral and a **second-stage caesarean**, with its impacted head, extensions, haemorrhage, bladder injury and a scar for life.

The one thing never to forget: *ventouse is safe only when it should be easy.* If the head is high, the position unknown, or there is no descent with a good pull, **stop and do a caesarean**. Most ventouse disasters (subgaleal haemorrhage, dead babies, torn mothers) come from **persisting** when the signs said stop.

Definitions

Term	Meaning
Assisted (operative / instrumental) vaginal birth	Vaginal birth assisted by vacuum or forceps
Vacuum extractor / ventouse	A cup (soft or rigid) attached to a vacuum source and traction handle
Flexion point	The point on the sagittal suture **about 3 cm anterior to the posterior fontanelle**. Traction over it flexes the head
Chignon	The artificial caput formed by scalp sucked into the cup. It resolves within hours
Pop-off / detachment	The cup losing its seal and coming off during traction
Pull	One episode of traction with one contraction
Station	The level of the **leading bony point of the skull** relative to the ischial spines (0 = at the spines; +1, +2 cm below)
Fifths palpable	How many fifths of the fetal head can be felt above the pelvic brim abdominally (5/5 to 0/5). **Always assessed before an instrumental delivery**

Why it matters in Uganda

- The WHO lists **7 BEmONC signal functions**: parenteral antibiotics, parenteral uterotonics, parenteral anticonvulsants, manual removal of the placenta, removal of retained products (e.g. MVA), **assisted vaginal delivery**, and neonatal resuscitation. **Assisted vaginal delivery is consistently the least-performed signal function** in sub-Saharan African EmONC assessments, including Uganda's. Reasons include no functioning equipment, lack of training and confidence, fear of blame, and a culture of "refer for CS".
- In high-income countries 10–15% of births are assisted. In many Ugandan facilities the figure is **around 1% or less**, while second-stage CS rates rise Verify *Ugandan facility rate is an estimate; no national figure cited.*
- **Second-stage CS** is more dangerous than CS in the first stage: deeply impacted heads, **extensions into the uterine vessels**, PPH, bladder injury, fetal skull and limb injuries, and neonatal asphyxia. A well-conducted ventouse avoids these.
- In Uganda, trained **midwives** as well as doctors perform ventouse deliveries. The *Essential Maternal and Newborn Clinical Care Guidelines* and EmONC training (e.g. ALSO/ESMOE-style courses) include vacuum extraction as a core skill.

Indications

Category	Examples
Fetal	**Suspected fetal compromise** in the second stage (persistent bradycardia, late or complicated decelerations, thick meconium with abnormal FHR)
Maternal: delay in the second stage	**Nulliparous**: active second stage lasting >2 h (NICE/RCOG allow up to 3 h total with regional analgesia). **Multiparous**: >1 h active (up to 2 h with regional). Ugandan teaching: **>2 h primigravida, >1 h multipara**. WHO 2018 / Labour Care Guide: delay is **≥3 h (nulliparous) or ≥2 h (parous)**
Maternal: exhaustion	Exhausted and unable to push effectively
Maternal: medical need to shorten pushing	Cardiac disease (NYHA III–IV), severe hypertension or hypertensive crisis, cerebrovascular disease or aneurysm, myasthenia gravis, proliferative retinopathy, spinal cord injury. **Forceps** may be preferred if maternal effort must be avoided entirely, because ventouse relies on some maternal pushing

UGANDA CONTEXT

Ugandan women rarely have epidural analgesia, so the second stage is usually shorter. Delay is commonly defined in local teaching as **>1 hour of active pushing in a multipara and >2 hours in a primigravida**. This is stricter than the WHO 2018 intrapartum recommendations and Labour Care Guide (**≥3 h nulliparous, ≥2 h parous** before calling it delay). Practice varies, so say which definition you are using. For a distressed baby with a low head at an **HC III**, a skilled ventouse **on site** may be safer than a 2-hour referral for CS. But if the head is **2/5 or more palpable**, or there is 3+ moulding, **refer or do a CS**. Never try ventouse "just to see" before a referral.

2. Contraindications and Precautions

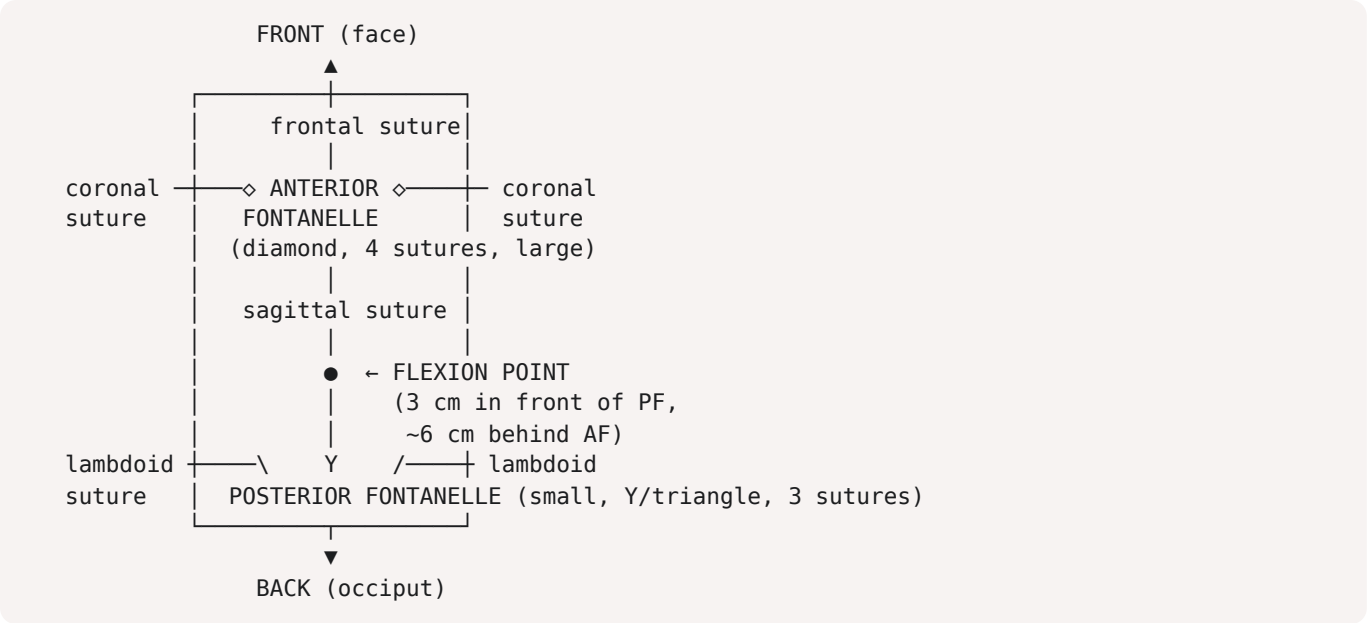
Absolute	Relative / caution
Cervix **not fully dilated** (except rare expert situations such as a second twin)	Gestation **32+0 to 36+0 weeks** (higher risk of cephalhaematoma and ICH)
Head **not engaged**: **≥2/5 palpable** abdominally or bony skull above the spines	**Mid-cavity** station (1/5 palpable, station 0/+1): trial in theatre by an experienced operator
Unknown position	**OP / deflexed** heads: use a posterior cup, expect a higher failure rate
Face, brow or breech presentation (and transverse lie)	**Macrosomia** (EFW >4 kg) and **BMI >30**: risk of failure and shoulder dystocia; trial in theatre
Gestation <32 weeks	**Recent fetal scalp blood sampling** or scalp electrode sites (bleeding)
Suspected **CPD** (3+ moulding, head high)	**Maternal HIV with unknown or unsuppressed viral load**, and hepatitis B/C: avoid where a safe alternative exists, because of scalp abrasions. **Don't let this delay an indicated birth**. Discuss with a senior
Fetal bleeding disorder (alloimmune thrombocytopenia, haemophilia) or **bone demineralisation** (osteogenesis imperfecta)	Previous failed vacuum in the same labour. Do not reapply
No consent, no back-up for CS, operator untrained	

Vacuum versus forceps: when forceps are the better instrument. Forceps are better for a face presentation (mento-anterior), the after-coming head of a breech, a preterm baby (32–36 weeks), when maternal pushing must be avoided, and when a large caput prevents cup adhesion. Forceps need specific skills that are increasingly rare in Uganda.

Precautions in every case: the bladder must be empty; analgesia must be adequate; a person trained in newborn resuscitation (HBB) must be present; and the team must be able to move to an **immediate CS** if the ventouse fails. Anticipate **shoulder dystocia** and **PPH**.

3. Relevant Anatomy and Normal Values

The fetal skull: landmarks you must feel



- **Posterior fontanelle**: small, **Y-shaped (3 sutures)**. When you feel it near the pubis, the position is **occipito-anterior (OA)**.
- **Anterior fontanelle (bregma)**: large, **diamond-shaped (4 sutures)**. If it is easily felt and central, the head is **deflexed** (often OP).
- If caput hides the sutures, feel for an **ear**: the **free edge of the pinna folds towards the occiput**. Where available, **intrapartum ultrasound** is more accurate than palpation for determining position (RCOG).

Why the flexion point matters

Cup placement	Presenting diameter	Result
Flexing median (centred over the flexion point, on the sagittal suture)	**Suboccipitobregmatic , 9.5 cm** (smallest)	Traction flexes the head, which autorotates and descends. This gives the highest success rate
Deflexing (too far forward, towards the anterior fontanelle)	Occipitofrontal, 11.5 cm	Traction extends the head, so a larger diameter presents, leading to failure, pop-offs and scalp trauma
Paramedian (off to one side of the sagittal suture)	Asynclitic head	Traction tilts the head, so the diameter increases and it fails to rotate

With a **6 cm cup** centred on the flexion point, the **posterior edge of the cup lies at the posterior fontanelle** and the **anterior edge lies about 3 cm behind the anterior fontanelle**. A cup edge that is on or near the anterior fontanelle means you are **too far forward**.

CLINICAL PEARL

In OP and deflexed positions the flexion point lies **far back, deep in the posterior vagina**. That is why a standard anterior cup (whose tubing comes out of the centre) cannot be placed correctly, and why **posterior cups** (Bird posterior cup, Kiwi OmniCup) have the tubing or traction cord coming out of the side or a groove. They are low-profile, so they can be slid right back to the flexion point.

The physics: why pressure and cup size matter

The maximum traction before the cup detaches is roughly **pressure × area**. A 6 cm cup at 0.8 kg/cm² can take about **20 kg of traction** before popping off. More pressure or a bigger cup allows more traction, but also more scalp damage. The limits (pressure, number of pulls, detachments, time) are there to **cap the total force** transmitted to the baby's scalp and brain.

Unit	Recommended working pressure
kg/cm²	**0.8** (range 0.6–0.8)
mmHg	**About 500–600**
kPa	**About 60–80**

Traction direction follows the **curve of Carus** (the pelvic axis): initially **downward and backward** until the occiput passes under the symphysis, then **horizontal**, then **upward** as the head crowns and extends (a "J-shaped" path). Always pull **perpendicular to the plane of the cup**. Oblique pulling breaks the seal and tears the scalp.

Classification of assisted vaginal birth (RCOG GTG 26, 2020)

Type	Definition	Comment
Outlet	Scalp visible at the introitus **without separating the labia**; the skull has reached the pelvic floor; sagittal suture in the AP diameter or ≤45° from it	Easiest and safest
Low	Leading point of the skull at **station +2 cm or below**, but not on the pelvic floor. Subdivided into ≤45° or >45° rotation	Standard ventouse territory
Mid	Head **1/5 palpable** abdominally; leading point **at station 0 to +1 cm**. Subdivided into ≤45° or >45° rotation	For experienced operators only; consider a **trial in theatre**
High	Head ≥2/5 palpable, or above the spines	**Not appropriate. Do a CS**

Normal values you need at the bedside

Item	Normal / required value
FHR	**110–160 bpm**; in the second stage check **every 5 minutes** (after each contraction) and between pulls
Contractions	Ideally **3–5 in 10 minutes**; tachysystole is >5 in 10 minutes
Head palpable abdominally	**0/5 or 1/5** required for ventouse
Bony station	**At or below the spines (0 or lower)**; +2 or below = low
Moulding	+1 (bones touching) or +2 (overlapping but reducible) acceptable; **+3 (irreducible) = stop**
Subpubic angle	≥2 fingers
Gestation	Ideally **≥34 weeks**; caution 32–36; **never <32 weeks**

Item	Normal / required value
Term neonatal blood volume	About **80–90 mL/kg** (about 280 mL for a 3.2 kg baby)

4. Equipment and Preparation

Equipment

Cup	Features	Best for	Cautions
Soft cup (silicone/silastic, e.g. Silc cup, Kiwi ProCup)	Bell or funnel shaped; pliable; conforms to the head	**OA, outlet and low** deliveries	**Higher failure** in OP, deflexed or large caput. **Less scalp trauma**
Rigid metal cup: anterior (Malmström, Bird anterior)	Stainless steel, mushroom shaped, 40/50/60 mm; separate chain or cord; **forms a chignon**	OA positions; outlet, low and mid	More scalp trauma (chignon, cephalhaematoma); must be reusable and sterilised
Rigid metal cup: posterior (Bird posterior)	**Tubing attached at the side**, separate traction chain	**OP and OT, deflexed heads** (reaches the flexion point)	Skilled operators only
Kiwi OmniCup (rigid plastic, disposable)	Low-profile "mushroom" cup, **integral hand pump (PalmPump) with a pressure gauge**, flexible stem; **traction force indicator** in some models	**All positions**, including OP (the stem lies in a groove, allowing a posterior application)	Single use; cost; the green zone on the gauge indicates working pressure
Pump	Hand pump or electric suction with a **gauge**		Check the gauge, tubing and seals **before** starting

Other equipment: sterile gloves, gown and PPE (eye protection); a delivery set; lubricant/antiseptic; a urinary catheter (in-out); local anaesthetic (**lidocaine 1%** for perineal infiltration or pudendal block) with syringe and needle; episiotomy scissors and suture materials; **oxytocin 10 IU** drawn up; a **newborn resuscitation set** (bag and mask, suction, warm towels) and a person trained in HBB; a documentation proforma.

> **UGANDA CONTEXT**
> Many Ugandan hospitals have **metal Malmström or Bird cups** (with hand or electric pumps) or silicone cups, and increasingly the disposable **Kiwi OmniCup** (hand-held, self-contained pump), supplied through partners. Check before you need it: cracked tubing and a dead pump are common problems.

Step 0: the first 5 minutes

The ventouse workup is a **bedside decision made in minutes**: is there a real indication, are **all the prerequisites** met, **where** should it be done (delivery room or trial in theatre), and is everything ready if it fails? Then, after the birth, there is a **post-procedure check** of mother and baby (section 9).

When you are called to a woman in the second stage with delay or an abnormal FHR:

1. **Mother**: pulse, BP, temperature, RR, colour, conscious level. **Collapse, shock, a hard tender uterus or heavy bleeding** point to **abruption or uterine rupture**, not a simple delay. That is a **Category 1 CS** or laparotomy, not a ventouse.
2. **Baby**: listen to the **FHR for a full minute straight after a contraction** (normal **110–160 bpm**; in the second stage check it **every 5 minutes**). A **prolonged bradycardia** needs birth **now**, by whichever route is fastest and safe.

3. **Call for help at once**: the most experienced operator available, a midwife, and a **person trained in newborn resuscitation (HBB)** with a bag and mask ready.
4. **Quick fixes while help comes**: turn her onto her **left side**, **stop oxytocin** if the FHR is abnormal or there is **tachysystole** (>5 contractions in 10 minutes), give IV fluids if she is dehydrated, and **empty her bladder**.
5. **Get the equipment out and checked in parallel** (cup, pump, tubing, gauge), plus lidocaine, a delivery set, oxytocin 10 IU drawn up and the resuscitation corner ready.
6. **Tell theatre early** if there is any doubt: a failed ventouse becomes a difficult second-stage CS with an impacted head.

Focused history: what to ask before deciding, and why

Question	Why it matters
Time of full dilatation and time active pushing started	Defines delay: **≥3 h (nulliparous) or ≥2 h (parous)** in the WHO 2018 / Labour Care Guide; common Ugandan ward teaching uses **>2 h primigravida and >1 h multipara** (longer with an epidural)
Parity and previous births (size of previous babies, previous instrumental delivery or CS)	Multiparas usually deliver quickly. Previous big babies suggest the pelvis is proven. A previous CS means a scar (short second stage, and ventouse is acceptable)
Gestational age	**<32 weeks is a contraindication**; 32–36 weeks calls for caution
Duration of the first stage and of pushing	A long first stage plus a long second stage suggests **CPD or malposition**, so ventouse is more likely to fail
Partograph: rate of dilatation, descent (fifths), moulding trend	Slow descent with **increasing moulding** means **obstruction**, and ventouse is dangerous
Contractions and oxytocin augmentation	Weak contractions ("power") may be the problem that a ventouse (with pushing) solves, or may be correctable first (oxytocin in a multipara needs great care)
Liquor colour and FHR pattern	Sets the urgency. A fetal indication shortens the decision time
HIV status and viral load; hepatitis B	Scalp abrasions; consider alternatives if the VL is unknown or unsuppressed; neonatal prophylaxis. It must not delay an indicated birth
Diabetes, obesity, estimated fetal weight, a previous shoulder dystocia	**Shoulder dystocia** risk. Think "trial in theatre"
Medical conditions (cardiac, severe pre-eclampsia)	Changes the indication and the choice of instrument
Bleeding disorders in the family; recent fetal scalp procedures	Contraindications
When she last passed urine	A full bladder obstructs descent and is injured by traction

Prerequisites checklist (RCOG)

Area	Requirement
Mother	Clear **explanation and verbal consent** (written for a trial in theatre); **analgesia** adequate (perineal infiltration or **pudendal block** for low or outlet; regional or spinal for mid-cavity or rotational); **bladder empty** (in-out catheter; deflate an indwelling balloon); **lithotomy** or dorsal position; **aseptic technique**
Fetus	**Cephalic**; **position known**; **head ≤1/5 palpable abdominally**; **bony station at or below the spines**; **moulding and caput** assessed (3+ moulding means stop); gestation ≥34 weeks ideally (never <32)
Birth canal	**Cervix fully dilated**; **membranes ruptured**; pelvis clinically adequate

Area	Requirement
Staff and set-up	Trained operator (or supervised trainee); equipment checked; **newborn resuscitator** present; ability to move to **immediate CS** if it fails; anticipation of **shoulder dystocia and PPH**

Consent, positioning and exposure

- **Consent**: explain to the woman (and her companion) what will happen and why, the alternatives (continued pushing, CS), and the main risks (failure, perineal tears, scalp marks, rarely serious scalp bleeding in the baby). Verbal consent in the delivery room; **written consent for a trial in theatre**. Tell her she will need to **push with each contraction**.
- **Position**: lithotomy or dorsal with the buttocks at the edge of the bed. Ensure good light.
- **Exposure and privacy**: expose only the abdomen and perineum; keep a chaperone or companion present; drape.
- **Decide the place**: **delivery room** (outlet or low, OA, good chance of success) or **trial in theatre** (mid-cavity, OP, 1/5 palpable, EFW >4 kg, BMI >30, prolonged labour). A trial in theatre lets you move to CS within minutes.

5. Procedure Step by Step

A. Focused examination before applying the cup (Hutchison order)

Do the examination **in this order** and write each finding down. **Every prerequisite must be met.**

1. **Introduction and consent**: introduce yourself, confirm her identity, explain, gain consent; chaperone present.
2. **Hand hygiene**.
3. **Position and exposure**: dorsal, abdomen exposed, then lithotomy for the VE.
4. **General inspection**: well or unwell, exhaustion, dehydration, pallor, distress; **vitals** (pulse, BP, temperature, RR). Shock, a hard tender uterus or heavy bleeding means abruption or rupture, not a ventouse.
5. **Abdomen (always before the VE)**:
   - **Inspection**: shape, scars (previous CS), **Bandl's ring** (obstruction), a distended bladder.
   - **Fundal height and lie**: longitudinal; estimated fetal size.
   - **Presentation**: cephalic.
   - **Engagement: fifths of the head palpable**: **0/5 or 1/5 required**. **2/5 or more = the head is not engaged: no ventouse.**
   - **Contractions**: frequency and strength (ideally **3–5 in 10 minutes**).
   - **Bladder**: palpable?
   - **Auscultation of the fetal heart** after a contraction (Pinard or Doppler).
6. **Vaginal examination**:

Item	What you need	Stop if
Cervix	**Fully dilated**, no rim	Any cervix felt around the head
Membranes	**Ruptured** (or perform ARM)	
Station of the bony skull (not the caput)	**At or below the ischial spines (0 or lower)**; +2 or below = low	Above the spines

Item	What you need	Stop if
Position	**Known exactly**: follow the sagittal suture to a fontanelle. **Posterior fontanelle** (small, Y-shaped, 3 sutures) near the pubis = **OA**; **anterior fontanelle** (large, diamond, 4 sutures) near the pubis = **OP**. If caput hides the sutures, feel for an **ear** (the free edge of the pinna points to the occiput) or use **ultrasound**	Position unknown
Moulding	+1 (bones touching) or +2 (overlapping but reducible)	**+3 (overlapping, irreducible)**, especially with the head ≥1/5 palpable: **CPD**
Caput	Small (graded + to +++)	Large caput reaching the introitus while the head is still palpable abdominally: **the caput is lying**
Asynclitism	Sagittal suture central	Marked asynclitism: higher failure rate
Pelvis	Adequate sacral curve, spines not prominent, subpubic angle ≥2 fingers	Contracted pelvis
Liquor	Clear	Thick meconium or blood: urgency; resuscitator present

7. **Put fifths and station together**: they must agree (table in section 6). If they don't, **trust the abdomen**.

B. Bedside checks before and during

Test	When	What it tells you
FHR by Pinard or Doppler	**Every 5 minutes** (after each contraction) in the second stage; **between pulls**	Urgency; a deteriorating FHR with birth not imminent means stop and go to CS
Partograph review	Before deciding	Rate of dilatation and descent, moulding trend, time in the second stage
In-out catheter	Before every application	Empties the bladder; note the volume. **Blood-stained urine** suggests obstruction
Urine dipstick	If there is time	**Ketones** (dehydration, exhaustion), protein (pre-eclampsia), blood (obstruction)
HemoCue Hb	If she looks pale or is known to be anaemic	Readiness for PPH (group and cross-match if low)
Intrapartum ultrasound (where available and you are trained)	Position uncertain	Fetal orbits and midline echo show the position more reliably than palpation
Equipment check	Before insertion	Occlude the cup on a gloved palm: the gauge must reach working pressure and hold
Contraction count	Before and during	Pull only with contractions; weak contractions reduce success

C. The procedure

1. **Explain** to the woman (and her companion) what will happen and why, and get **consent**. Tell her she will need to **push with each contraction**.
2. **Position**: lithotomy or dorsal with the buttocks at the edge of the bed. Ensure good light.
3. **PPE and asepsis**: hand hygiene, sterile gloves, apron, eye protection. Clean the vulva and perineum and drape.

4. **Empty the bladder** with an in-out catheter.
5. **Re-examine** (abdominally and vaginally): fifths palpable, dilatation, station of the bony skull, **position** (fontanelles and sutures), caput and moulding. **If any prerequisite is not met, stop.**
6. **Analgesia**: perineal infiltration with **lidocaine 1% (up to about 20 mL)**, or a **pudendal block** (lidocaine 1%, 10 mL each side, max total **about 3–4.5 mg/kg without adrenaline**).
7. **Check the equipment**: connect the cup to the pump. Test for vacuum by occluding the cup against a gloved palm.
8. **Insert the cup**: separate the labia, compress the cup (soft) or insert it sideways (rigid) with pressure on the posterior vaginal wall, and slide it onto the scalp.
9. **Locate the flexion point**: place the **centre of the cup over the sagittal suture about 3 cm in front of the posterior fontanelle**. Check that the anterior edge is about 3 cm from the anterior fontanelle.
10. **Create initial vacuum** (about **0.2 kg/cm²**, about 150 mmHg). **Sweep a finger around the whole cup circumference** to make sure no **cervix or vaginal wall** is trapped under it (trapping causes tears and failure).
11. **Increase to working pressure** (about **0.8 kg/cm²**, 500–600 mmHg, or the green zone). Rapid build-up over 1–2 minutes is as effective as the old stepwise method.
12. **Traction with the contraction and maternal push**:
    - **Dominant hand** pulls on the handle, perpendicular to the cup, along the pelvic curve: **downward first, then horizontal, then upward** as the head crowns.
    - **Non-dominant hand**: the **thumb on the cup** and the **index finger on the scalp** next to it (the "finger-thumb" position) to feel for **descent** and for early **cup slippage**, and to keep the head flexed.
    - **Do not rock or twist** the cup. The head **autorotates** as it descends.
13. **Between contractions**: stop pulling, check the FHR, and reassess descent. (With metal cups some operators reduce the vacuum to about 0.2 kg/cm²; with the Kiwi, follow the manufacturer.)
14. **Assess each pull**: there must be **descent with every pull**. Expect the head to be **crowning or delivered within 3 pulls**.
15. **Episiotomy**: **not routine** with ventouse. Consider a **mediolateral** episiotomy (at about 60° from the midline at crowning) if the perineum is tight and the baby needs rapid delivery, or if tearing seems imminent.
16. **Delivery of the head**: guard the perineum. When the **jaw is reachable**, **release the vacuum** and remove the cup. Deliver the shoulders and body normally, and **anticipate shoulder dystocia** (call for help early).
17. **Third stage**: **oxytocin 10 IU IM** (active management), controlled cord traction, uterine massage. Watch for **PPH**.
18. **Inspect** the vagina, cervix and perineum for tears and repair them. Perform a **PR exam** to exclude an anal sphincter injury (OASI), before and after the repair.
19. **Examine the baby**: scalp (chignon, abrasions, lacerations, swelling), head circumference, and **vitamin K 1 mg IM** (routine in Uganda). Hand over for **enhanced neonatal observations**.
20. **Document** (section 9) and **debrief** the woman.

D. When to abandon

RED FLAG

Stop and proceed to caesarean section if:

- There is **no descent with a pull** that was correctly applied and done with a contraction.
- Delivery is **not imminent after 3 pulls** (the head should be crowning or delivered).
- The cup **detaches twice** (some guidance allows up to 3 detachments). **Do not keep reapplying.**
- The total cup application time exceeds about **15–20 minutes**. The WHO *Managing Complications in Pregnancy and Childbirth* manual allows up to 30 minutes, but modern guidance (RCOG) is stricter.
- **Fetal condition deteriorates** and the birth is not imminent.

Avoid sequential instruments (ventouse, then forceps). This raises the risk of intracranial haemorrhage and other neonatal trauma. In Uganda, where forceps skills are scarce, the fallback is **CS**. Tell theatre early, and warn them that the head may be deeply impacted.

6. Findings and Interpretation

Fifths and station together (they must agree)

Abdomen	VE (bony skull)	Classification (RCOG)	Where and who
0/5	Scalp visible without parting the labia	**Outlet**	Delivery room; trained midwife or doctor
0/5	**+2 or lower**, not on the pelvic floor	**Low**	Delivery room; trained operator
1/5	**0 to +1**	**Mid-cavity**	**Trial in theatre** by an experienced operator, ready for immediate CS
≥2/5	Above the spines (or caput low but skull high)	**High**	**No ventouse. CS**

If the abdomen and the VE disagree (e.g. "station +2" but 2/5 palpable), **trust the abdomen**: a big caput has misled the VE.

What each examination finding means

Finding	Meaning	Action
Head 0/5 or 1/5 palpable, bony skull at or below the spines	Engaged; ventouse possible	Proceed if all other prerequisites are met
Head ≥2/5 palpable	Not engaged	No ventouse; CS
Small Y-shaped fontanelle anteriorly	OA	Standard cup placement
Large diamond fontanelle anteriorly, or easily felt and central	OP or deflexed	Posterior cup (Kiwi, Bird posterior); trial in theatre; higher failure
Moulding +3 with head ≥1/5 palpable	CPD	No ventouse; CS
Large caput, oedematous vulva, prolonged labour	Obstructed labour	CS
Marked asynclitism	Paramedian presentation	Higher failure; reassess
Cervical rim felt	Not fully dilated	No ventouse

Finding	Meaning	Action
Thick meconium, abnormal FHR	Fetal compromise	Urgency; resuscitator present
Blood-stained urine on catheterisation	Obstruction (bladder compression)	Reconsider; CS more likely
Bandl's ring, haematuria, tender lower segment	Obstructed labour, impending rupture	No ventouse; urgent CS

Laboratory and imaging

Most ventouse decisions are made **without lab results**. Do not delay a clearly indicated ventouse for tests. The table shows what matters before and, especially, after the birth.

Test	When	Expected / abnormal finding	How it changes management
Maternal Hb (HemoCue at HC III/IV; FBC at hospital)	Before, if anaemic; after, if PPH	Hb <11 g/dL anaemia; **<7 g/dL severe**	Group and cross-match; low threshold to treat PPH; iron afterwards
Blood group and cross-match (HC IV with blood bank; hospitals)	**Trial in theatre**, anaemia, prolonged labour, PPH		Blood ready for a failed ventouse and CS, or for PPH
HIV rapid test (all levels)	Status unknown	Reactive	Infant prophylaxis and EID; ART for the mother. **Do not delay an indicated birth** for the result
Intrapartum ultrasound (some hospitals; rarely at HC III/IV)	Position unclear on VE	OA, OT or OP; fetal size	Choice of cup (posterior cup for OP) or CS
Cord blood gas (few Ugandan units)	Abnormal FHR before birth, or a depressed baby	pH <7.0 or a large base deficit = significant acidosis	Neonatal care; documentation
Neonatal Hb / PCV (HemoCue or lab; hospitals)	**Any enlarging scalp swelling**, pallor or tachycardia after ventouse; repeat serially	**Falling Hb/PCV**	Suspected **subgaleal haemorrhage**: normal saline **10–20 mL/kg**, then **blood transfusion**
Neonatal clotting (PT, APTT, platelets) (RRH/national)	Suspected subgaleal haemorrhage	Prolonged times, low platelets (consumption)	FFP, platelets, vitamin K
Neonatal bilirubin (hospitals; transcutaneous or serum)	Cephalhaematoma or subgaleal bleed, jaundice	Rising bilirubin from breakdown of the haematoma	Phototherapy
Cranial ultrasound / CT (ultrasound at many hospitals; CT at RRH/national or private)	Seizures, apnoea, bulging fontanelle, altered tone	**Intracranial haemorrhage**	Neonatal intensive care; neurosurgical advice
Bladder volume (bladder scan or catheter, all levels)	She has not voided within about **6 hours**	Large residual	Indwelling catheter for 24–48 h

Scalp swellings after birth

CLINICAL PEARL

Cephalhaematoma vs subgaleal haemorrhage vs caput succedaneum is a classic viva.

- **Caput**: scalp oedema, **crosses sutures**, present **at birth**, soft and pitting, gone in 1–2 days.
- **Cephalhaematoma**: **subperiosteal**, does **not** cross sutures, appears **hours later**, resolves in weeks.
- **Subgaleal**: **subaponeurotic**, **crosses sutures**, **enlarges** over hours, can cause **shock**, and is life-threatening.

The loose space under the galea can hold **several hundred mL**, **most of a term baby's blood volume** (about 80–90 mL/kg, so about 280 mL for a 3.2 kg baby).

Swelling	Layer	Crosses sutures?	Timing	Significance
Chignon	Scalp sucked into the cup	Limited to the cup site	At birth	Normal; resolves in **12–24 h**
Caput succedaneum	Scalp oedema	Yes	At birth	Benign; gone in 1–2 days
Cephalhaematoma	Subperiosteal	No	Hours after birth	Resolves over weeks; jaundice, anaemia
Subgaleal haemorrhage	Subaponeurotic	Yes, orbits to nape	Enlarges over hours	Hypovolaemic shock; emergency

7. Clinical Acumen

Red flags

RED FLAG

- **Head 2/5 or more palpable abdominally**, even if the caput is "at the introitus": the head is not engaged. **No ventouse.**
- **+3 moulding, big caput, oedematous vulva, prolonged labour**: obstructed labour. CS.
- **Can't determine the position** (and no ultrasound): don't apply a cup blindly. Call a senior.
- **No descent with the first well-applied pull** in a multipara, or with the second in a primigravida: likely failure. Reassess seriously.
- **After delivery**: a baby with **pallor, tachycardia, poor tone, a boggy fluctuant scalp swelling crossing the suture lines, or a rising head circumference** has **subgaleal haemorrhage** until proven otherwise.
- **After delivery**: maternal **PPH**, perineal pain out of proportion (haematoma), inability to void (retention).

Thinking like a clinician

1. **"Is this a delay of power, passenger or passage?"**
   - **Power** (weak contractions, exhaustion) with a low head in good position: a ventouse works well.
   - **Passenger** (OP, deflexed, asynclitic, big baby): ventouse may work with a posterior cup and correct placement, but failure is more likely.
   - **Passage** (CPD, contracted pelvis): **ventouse will fail and harm**. CS.
2. **Decide the place**: in the **delivery room** (outlet or low, OA, a good chance of success) or as a **trial in theatre** (mid-cavity, OP, 1/5 palpable, EFW >4 kg, BMI >30, prolonged labour). A trial in theatre lets you move to CS

within minutes.

3. **Decide the instrument**: **ventouse** is first-line for most Ugandan operators because it needs less analgesia, causes less maternal trauma, and midwives are trained in it. **Forceps** are better for a face presentation (mento-anterior), the after-coming head of a breech, a preterm baby (32–36 weeks), when maternal pushing must be avoided, and when there is a large caput preventing cup adhesion.
4. **If X then think Y**:
    - Caput at the introitus but the head 2/5 palpable: the **caput is lying to you**. The skull is high. CS.
    - A pop-off on the first pull: **check the application** (flexion point? tissue trapped? wrong direction of pull?). One reapplication is allowed. A second pop-off means stop.
    - The head descends but doesn't rotate from OT or OP: flexion is probably poor. Was the cup too anterior? Consider stopping.
    - The baby is delivered but "the scalp looks swollen all over, not just the chignon": **think subgaleal haemorrhage** and start hourly observations and head measurements.
    - Collapse, shock, a hard tender uterus or heavy bleeding in the second stage: **abruption or uterine rupture**, not a delay. Category 1 CS or laparotomy.
5. **When to stop or escalate**: any prerequisite not met; no descent with a correct pull; 3 pulls without imminent birth; 2 detachments; 15–20 minutes; fetal deterioration with birth not imminent. Escalate early to the senior and theatre rather than late.
6. **Common student mistakes**:
    - Relying on VE station while ignoring abdominal fifths.
    - Placing the cup where it is easiest (just inside the introitus, often over the anterior fontanelle) rather than at the flexion point.
    - Pulling between contractions, or pulling upward too early.
    - Twisting the cup to "rotate" the head.
    - Persisting after pop-offs.
    - Forgetting the bladder, the paediatrician (or HBB-trained person), oxytocin for the third stage, and the PR exam after the repair.

Putting it together: worked examples

Model decision summary: "Mrs A., P0+0 at 39+5 weeks, **fully dilated for 3 h, pushing for 2 h 15 min**, exhausted, FHR 150 with early decelerations. Head **0/5 palpable**, bony station **+2**, **LOA**, caput +, moulding +1, clear liquor, bladder emptied. Impression: **prolonged second stage with maternal exhaustion; low-cavity, OA; all prerequisites met**. Plan: **ventouse in the delivery room** after consent and perineal infiltration, with a newborn resuscitator present, theatre informed as back-up, and oxytocin 10 IU IM for the third stage."

Worked example 1: "Go now" (fetal indication). A P2 has been pushing for **1 h 15 min**. The FHR drops to **90 bpm for 3 minutes** and does not recover. On examination the head is **0/5**, the bony skull is **+3** and just visible with the labia parted, **LOA**, moulding +1. *Interpretation:* **suspected fetal compromise** with a **low (nearly outlet) OA head**; every prerequisite is met, and a CS would take far longer. *Action:* call for help and the resuscitator, empty the bladder, infiltrate the perineum, and do a **ventouse now in the delivery room**, expecting delivery in **1–2 pulls**. If there is **no descent with the first correct pull**, stop and move to Category 1 CS.

Worked example 2: "Trial in theatre or CS" (reading the findings). A primigravida at 40 weeks has pushed for **2 h 30 min** after a 16-hour first stage augmented with oxytocin. The head is **1/5 palpable**; VE: bony station **0 to +1**, **caput ++**, **moulding +2**, the anterior fontanelle is felt behind the pubis, so the position is **direct OP**. EFW 3.9 kg. *Interpretation:* a **mid-cavity OP** head after a **long labour** (both passenger and possibly passage factors): the failure risk is **high**. *Action:* this is **not** a delivery-room ventouse. It is either a **trial in theatre** by an experienced operator with a **posterior cup** (Kiwi OmniCup or Bird posterior), with consent for CS and the anaesthetist ready, or a **CS** directly. If the head were **2/5 palpable** or moulding **+3**, the answer would be **CS only**.

8. Complications and Troubleshooting

Immediate maternal aftercare

Issue	Action
Third stage and PPH prevention	**Oxytocin 10 IU IM** (or IV slowly) immediately after birth. Controlled cord traction. Check the uterus every 15 min for the first hour. Be ready to treat PPH (bimanual compression, oxytocin infusion 20 IU in 1 L, **tranexamic acid 1 g IV over 10 min**, misoprostol 800 µg sublingual)
Genital tract trauma	Inspect the cervix, vagina and perineum. Repair tears and episiotomy with **2-0 Vicryl Rapide** (or chromic catgut). **PR exam** before and after repair to detect **OASI**. Third/fourth-degree tears go to an experienced operator in theatre
Antibiotic prophylaxis	RCOG (ANODE trial): a **single IV dose of amoxicillin-clavulanate (1.2 g)** after assisted vaginal birth halves maternal infection. If it is unavailable, follow the unit protocol (e.g. a single dose of **ampicillin 2 g IV ± metronidazole 500 mg IV**). Give a full course only for established infection or third/fourth-degree tears per protocol
Analgesia	**Paracetamol 1 g oral 6-hourly** + **diclofenac 50 mg oral 8-hourly** (or ibuprofen 400 mg 8-hourly) for 3–5 days, unless contraindicated (severe pre-eclampsia with renal impairment, asthma, PPH with oliguria)
Bladder care	High risk of **urinary retention** (swelling, pain, regional block). Ensure she **voids within about 6 hours** and check the volume. If she can't void or the bladder is palpable, catheterise (and consider an indwelling catheter for 24–48 h if the residual is large)
VTE assessment	Assisted vaginal birth is a VTE risk factor. Add other risks (age >35, obesity, PPH, prolonged labour, immobility, sickle cell). Give **enoxaparin 40 mg SC daily** if the score warrants it, plus early mobilisation
Debrief	Explain why the ventouse was needed, how it went, what the chignon is, what to watch for in the baby, and the implications for the next birth (**most women who have had an assisted delivery have a spontaneous vaginal birth next time**)

UGANDA CONTEXT

The ANODE single-dose amoxicillin-clavulanate recommendation comes from RCOG/UK practice; Ugandan units often use ampicillin ± metronidazole by local protocol, and amoxicillin-clavulanate IV may not be stocked Verify *whether UCG 2023 or the Uganda EMHSLU lists IV amoxicillin-clavulanate for this indication*.

Neonatal complications

Complication	Features	Management
Chignon	Soft, circumscribed swelling exactly where the cup was	Reassure. It **resolves in hours (12–24 h)**
Scalp abrasions / lacerations	From twisting, prolonged traction or rotation	Clean; keep dry; they usually heal. Watch for infection
Cephalhaematoma	**Subperiosteal** bleed, **limited by suture lines** (does not cross them), usually over a parietal bone, appears **hours after birth**, firm and fluctuant	**No aspiration** (infection risk). It resolves over **weeks to months**. Watch for **jaundice** (breakdown of blood) and anaemia. Rarely calcifies

Complication	Features	Management
Subgaleal (subaponeurotic) haemorrhage	Bleeding between the **epicranial aponeurosis (galea) and the periosteum**. **Crosses suture lines**, spreading from the orbits to the nape and around the ears. A boggy, fluctuant, shifting swelling; **pitting of the scalp**; ears pushed forward; **increasing head circumference** (about **1 cm increase ≈ 40 mL blood**); **pallor, tachycardia, hypotonia, hypovolaemic shock**; later DIC	**Emergency**: call the paediatrician or neonatal team. **Hourly** obs (HR, RR, colour, tone, cap refill), **measure head circumference every hour**, Hb/PCV, coagulation screen. **Volume**: normal saline **10–20 mL/kg** IV, then **blood transfusion** (O-negative or cross-matched), FFP for coagulopathy, **vitamin K 1 mg IM/IV**. Transfer to NICU. Mortality can reach 12–25%
Intracranial haemorrhage	Apnoea, seizures, poor feeding, bulging fontanelle, altered tone	Neonatal team; cranial ultrasound or CT; supportive care
Retinal haemorrhages	Common (vacuum > forceps), asymptomatic	Resolve spontaneously
Neonatal jaundice	Resorption of haematoma	Bilirubin monitoring; phototherapy
Shoulder dystocia and brachial plexus injury	Associated with macrosomia and instrumental birth	Anticipate: call for help, McRoberts, suprapubic pressure (HELPERR). Follow up the brachial plexus

Routine neonatal observation after every ventouse: examine the scalp and measure the head circumference at birth. Do **regular observations (at least hourly for the first few hours, then 2–4-hourly for 12–24 h)** of HR, RR, colour, tone and feeding, and of the **size and consistency of scalp swelling**. Teach the mother and ward staff what to look for. Give **vitamin K 1 mg IM** to every baby.

Troubleshooting during the procedure

Problem	Likely cause	Fix
Cup will not hold vacuum on the palm test	Cracked tubing, worn seal, dead pump	Replace before insertion; never start with faulty equipment
Pop-off on the first pull	Wrong placement (too anterior), trapped tissue, oblique pull, large caput	Check the flexion point, finger sweep, pull perpendicular; **one reapplication** only
Second pop-off	Failing ventouse	**Stop**; CS
No descent with a correct pull	Malposition, CPD	**Stop**; CS
Head descends but does not rotate from OT/OP	Poor flexion; cup too anterior	Consider stopping; do not twist the cup
Large caput prevents cup adhesion	Prolonged labour	Consider forceps (if skilled) or CS
Shoulders do not deliver	Shoulder dystocia	Call for help; HELPERR (McRoberts, suprapubic pressure)

Failed ventouse: what next?

1. **Stop.** Do not reapply the cup or switch to forceps unless the head is at the outlet and an experienced operator judges it safe.
2. **Call the senior and theatre**: this is usually a **Category 1 or 2 CS** depending on the FHR.
3. Keep monitoring the FHR. Stop any oxytocin if there is fetal distress. Consider **acute tocolysis** (terbutaline 0.25 mg SC) for an abnormal FHR while preparing.
4. **Warn the team of an impacted head**: the assistant may need to **disimpact the head vaginally** ("push"), or the surgeon may deliver by reverse breech extraction ("pull"). Expect **extensions** and **PPH**.
5. **Neonatal team present at the CS**: failed ventouse followed by CS carries a high risk of neonatal injury and asphyxia.
6. **Document** everything and **debrief** the woman and the team.

9. Documentation and Presenting Findings

Documentation (every ventouse)

- **Indication** and **urgency**.
- **Prerequisites**: abdominal fifths, cervical dilatation, **station**, **position**, caput and moulding, liquor, FHR.
- **Consent** (verbal or written) and **analgesia**.
- **Instrument and cup type**, **operator** (and supervisor).
- **Time of cup application** and **time of delivery**. **Number of pulls** and **number of detachments**.
- Episiotomy or tears and their repair. **Estimated blood loss**.
- **Baby**: sex, weight, Apgars, resuscitation, **scalp findings**, head circumference, cord gases if available.
- **Plan**: bladder care, VTE, antibiotics, neonatal observations, and debrief.

Post-procedure checks (write them in the notes)

Mother	Baby
Oxytocin 10 IU IM given; uterus firm; **EBL** recorded; fundus and lochia every 15 min for the first hour	Apgars; resuscitation needed?
Cervix, vagina and perineum **inspected**; **PR exam** before and after repair (OASI)	**Scalp** examined: chignon (normal), abrasions, lacerations, swelling
Voided within about 6 hours? Volume checked	**Head circumference at birth**, then observations (HR, RR, colour, tone, **scalp swelling**) **at least hourly for the first few hours**, then 2–4-hourly for 12–24 h
Antibiotic prophylaxis given (single dose, per unit protocol)	**Vitamin K 1 mg IM**
Analgesia; **VTE risk assessed**	Mother taught to report pallor, poor feeding or a growing swelling. **1 cm increase in head circumference ≈ 40 mL of blood**
Documentation: pulls, detachments, application time; **debrief**	Enhanced neonatal observations handed over to the ward

Model case presentation

> "Mrs K.S., a 23-year-old **primigravida at 39+2 weeks**, was delivered by **ventouse** at 03:40 for **prolonged second stage** (2 h 20 min of active pushing) with **maternal exhaustion** and a **non-reassuring FHR** (late decelerations). Before the procedure the head was **1/5 palpable**, the cervix fully dilated, the bony station

> **+2**, position **left occipito-anterior**, caput +, moulding +1, clear liquor. After consent, in-out catheterisation and perineal infiltration with 10 mL lidocaine 1%, a **Kiwi OmniCup** was applied at the flexion point. There were **2 pulls with descent and no detachments**, and the application time was 8 minutes. A live female, 3.3 kg, Apgars 6 and 9, was born, with a **chignon** only and head circumference 34 cm. She had a second-degree tear, repaired with 2-0 Vicryl Rapide; the PR exam was normal. EBL 400 mL. She received oxytocin 10 IU IM and a single dose of IV antibiotics. **Plan**: paracetamol and diclofenac, **void within 6 hours** with a volume check, VTE risk assessment, **neonatal observations and head circumference every hour for 4 hours then 4-hourly for 24 hours**, and a debrief today."

Note: in this model case the head was 1/5 palpable with station +2; by the RCOG table a 1/5 head is mid-cavity territory, so in a real presentation say why it was judged a low delivery (e.g. the head was 0/5 on re-examination after the catheter) or present it as a trial in theatre Verify *fifths and station in the model case do not quite match the RCOG low-cavity definition*.

10. OSCE Checklist

Station: "Perform a vacuum extraction on the manikin" (or "talk me through it")

Step	Done well when	Critical?
1. Introduction, indication, consent	Introduces self; explains the indication and procedure; obtains consent	Yes
2. Calls for help	Asks for a senior or midwife and a newborn resuscitator	Yes
3. States the prerequisites	Fully dilated, membranes ruptured, head ≤1/5 palpable, station at or below spines, position known, no 3+ moulding	Yes
4. Position and PPE	Lithotomy; hand hygiene; sterile gloves; apron; eye protection	No
5. Asepsis and bladder	Cleans and drapes; **empties the bladder**	Yes
6. Examination	Abdominal palpation (fifths) **before** the VE (dilatation, station, position, caput, moulding)	Yes
7. Analgesia	Gives or states perineal infiltration or pudendal block	No
8. Equipment check	Checks the cup, pump and vacuum on a gloved palm	No
9. Cup placement	Inserts the cup correctly and places it on the **flexion point** (3 cm anterior to the posterior fontanelle, on the sagittal suture)	Yes
10. Finger sweep	Creates low vacuum; **sweeps a finger around the cup** to exclude maternal tissue	Yes
11. Working pressure	Increases to about 0.8 kg/cm² / 600 mmHg / green zone	No
12. Traction	**Only with contractions** plus maternal effort, **perpendicular to the cup**, along the **pelvic curve**, with the finger-thumb technique	Yes
13. Descent	Checks for **descent with each pull**; does not rock or twist	Yes
14. Abandon criteria	States them: no descent; 3 pulls; 2 detachments; 15–20 min	Yes
15. Perineum	Considers episiotomy selectively; guards the perineum	No
16. Delivery	Releases the vacuum when the jaw is reachable; delivers the baby	No
17. Third stage	Active management (oxytocin 10 IU IM)	Yes
18. Genital tract	Inspects for tears; PR exam; repair	No
19. Baby	Examines the scalp; vitamin K; neonatal observation plan	Yes
20. Documentation	Documents fully and debriefs the mother	No

11. Key Points

High-yield points

- **Ventouse is safe only when it should be easy.** Stop when the signs say stop.
- Assisted vaginal delivery is a **BEmONC signal function** and the least performed in Uganda.
- **Prerequisites**: fully dilated, membranes ruptured, head **≤1/5 palpable** and **at or below the spines**, position known, no +3 moulding, empty bladder, consent, analgesia, trained operator, resuscitator, CS back-up.
- **Abdomen before VE**; if they disagree, **trust the abdomen**.
- Cup on the **flexion point** (3 cm in front of the posterior fontanelle) for a flexing median application.
- **0.8 kg/cm²**, pull **with contractions**, **perpendicular**, along the **J-shaped** pelvic curve; never twist.
- **3-2-1-20** stopping rules; no sequential instruments; fallback is CS.
- Vacuum: more failure, cephalhaematoma and retinal haemorrhage; forceps: more maternal perineal trauma.
- **Subgaleal haemorrhage** crosses sutures and can kill within hours: hourly scalp checks and head circumference.
- After every ventouse: **oxytocin 10 IU IM**, PR exam, void within 6 h, single-dose antibiotic, VTE assessment, vitamin K for the baby.

Numbers & doses

Item	Value
Head palpable abdominally	**0/5 or 1/5** (≥2/5 = no ventouse)
Bony station	**At or below the spines**; low = **+2 or below**; mid = **0 to +1**
Flexion point	On the sagittal suture **3 cm in front of the posterior fontanelle** (about **6 cm behind the anterior fontanelle**)
Diameter presented by a flexing median application	**Suboccipitobregmatic 9.5 cm** (deflexed: occipitofrontal 11.5 cm)
Initial vacuum (for the finger sweep)	About **0.2 kg/cm²** (about 150 mmHg)
Working pressure	About **0.8 kg/cm²** = **500–600 mmHg** = **60–80 kPa**
Traction before a 6 cm cup detaches	About **20 kg** at 0.8 kg/cm²
Stopping rules ("3-2-1-20")	**3** pulls without imminent birth, **2** detachments (some guidance allows 3), **1** correct pull without descent, **15–20 min** application time (WHO MCPC allows up to 30 min)
Gestation	Ideally **≥34 weeks**; caution 32–36; **never <32 weeks**
Second-stage delay	WHO 2018 / LCG: **≥3 h nulliparous, ≥2 h parous**; common Ugandan/NICE-style teaching: **>2 h primigravida, >1 h multipara** (longer with an epidural)
FHR in the second stage	**Every 5 minutes** (after each contraction); normal **110–160 bpm**
Perineal infiltration	**Lidocaine 1%**, up to about **20 mL**
Pudendal block	Lidocaine 1%, **10 mL each side**; maximum about **3–4.5 mg/kg** without adrenaline
Third stage	**Oxytocin 10 IU IM**
PPH treatment	Oxytocin infusion **20 IU in 1 L**; **tranexamic acid 1 g IV over 10 min**; misoprostol **800 µg sublingual**
Antibiotic after assisted birth (ANODE)	**Amoxicillin-clavulanate 1.2 g IV single dose** (or unit alternative: ampicillin 2 g IV ± metronidazole 500 mg IV)

Item	Value
Analgesia	Paracetamol **1 g 6-hourly** + diclofenac **50 mg 8-hourly** (or ibuprofen 400 mg 8-hourly), 3–5 days
VTE prophylaxis	Enoxaparin **40 mg SC daily** if the risk score warrants
Acute tocolysis	Terbutaline **0.25 mg SC**
Voiding after birth	Within about **6 hours**
Neonatal vitamin K	**1 mg IM**
Subgaleal haemorrhage	**1 cm rise in head circumference ≈ 40 mL of blood**; normal saline **10–20 mL/kg**, then blood; mortality up to 12–25%
Chignon	Resolves in **12–24 h**
Episiotomy (if needed)	**Mediolateral, about 60°** from the midline at crowning; not routine

Exam traps

- **"The caput is at the introitus, so the head is low."** No. **Check the fifths abdominally**; if 2/5 or more is palpable, the skull is high: CS.
- **"Try the ventouse before referring, just to see."** No. A ventouse is done only when **every prerequisite is met**; otherwise refer or do a CS.
- **"Twist the cup to rotate an OP head."** No. **Never rotate**; correct placement on the flexion point lets the head **autorotate**.
- **"Pull steadily between contractions to save time."** No. Pull **only with contractions and maternal pushing**.
- **"If the ventouse fails, switch to forceps."** Avoid **sequential instruments** (more intracranial haemorrhage); the usual fallback in Uganda is **CS**.
- **"Put the cup where it sits easily, just inside the introitus."** That is usually over the **anterior fontanelle** (deflexing). Place it on the **flexion point**.
- **"A cephalhaematoma crosses the suture lines."** No. A **cephalhaematoma stops at the sutures**; **caput and subgaleal haemorrhage cross them**.
- **"Aspirate a cephalhaematoma."** No. Leave it (infection risk); watch for jaundice.

Questions seniors ask

Viva questions

1. **Where is the flexion point and why does it matter?** On the sagittal suture about 3 cm in front of the posterior fontanelle. Traction there flexes the head so the smallest diameter (suboccipitobregmatic, 9.5 cm) presents.
2. **What pressure do you use?** About 0.8 kg/cm² (500–600 mmHg, about 80 kPa).
3. **Name two contraindications specific to vacuum rather than forceps.** Gestation <32 weeks (caution <36), and face presentation (forceps can deliver a mento-anterior face). Also fetal bleeding disorders.
4. **Compare vacuum and forceps.** Vacuum fails more often and causes more cephalhaematoma and retinal haemorrhage, but less maternal perineal trauma, and needs less analgesia. Forceps are more successful but cause more OASI and facial injuries.
5. **Distinguish caput, cephalhaematoma and subgaleal haemorrhage.** Caput crosses sutures and is present at birth; cephalhaematoma stops at sutures and appears hours later; subgaleal crosses sutures, enlarges over hours and can cause shock (section 6).
6. **Why empty the bladder?** A full bladder obstructs descent, is at risk of injury, and causes post-partum retention.

7. **Why is a sequential instrument delivery discouraged?** It causes more neonatal intracranial haemorrhage and trauma, and more maternal trauma.

On the ward round

1. **What must you check before applying a ventouse? Fully dilated**, **membranes ruptured**, head **≤1/5 palpable** with the bony skull **at or below the spines**, **position known**, moulding not +3, bladder empty, consent, analgesia, a trained operator, a newborn resuscitator and **CS back-up**.
2. **The VE says +2 but you can feel 2/5 of the head abdominally. What do you do? Trust the abdomen**: the caput is misleading you. No ventouse; CS (or a senior review).
3. **How do you tell OA from OP on VE?** Follow the sagittal suture: a **small Y-shaped posterior fontanelle** anteriorly means **OA**; the **large diamond anterior fontanelle** anteriorly means **OP**. If unsure, feel an ear or use ultrasound.
4. **When would you do a trial in theatre? Mid-cavity** (1/5 palpable, station 0 to +1), **OP**, **EFW >4 kg**, BMI >30, or a long labour, where failure is more likely and immediate CS must be possible.
5. **When do you abandon?** No descent with a correct pull; not imminent after **3 pulls**; **2 detachments**; about **15–20 minutes**; or fetal deterioration with birth not imminent.
6. **What observations does the baby need after a ventouse, and why?** Scalp and **head circumference** at birth, then **at least hourly** HR, RR, colour, tone and scalp swelling for the first few hours, to catch **subgaleal haemorrhage** early. This is the most important post-delivery observation.
7. **What maternal checks follow a ventouse?** Oxytocin 10 IU IM, EBL, inspection and **PR exam** (OASI), repair, **voiding within 6 hours**, a single antibiotic dose, analgesia, VTE assessment and a **debrief**.

Mnemonics

"A–J" of vacuum extraction (ALSO course):

Letter	Meaning
A	**A**sk for help; **A**ddress the patient (consent); **A**naesthesia adequate
B	**B**ladder empty
C	**C**ervix fully dilated (and membranes ruptured)
D	**D**etermine position and station (fifths!); think shoulder **D**ystocia
E	**E**quipment ready and checked; **E**xtractor tested
F	**F**ontanelle: cup over the **F**lexion point (3 cm in front of the posterior fontanelle)
G	**G**entle traction with contractions, along the pelvic curve
H	**H**alt: stop between contractions; **H**alt the procedure if no descent, 3 pulls, 2 pop-offs, 15–20 min
I	**I**ncision: evaluate the need for episiotomy
J	**J**aw reachable: remove the cup

Prerequisites "FORCEPS": **F**ully dilated; **O**cciput position known; **R**uptured membranes; **C**ephalic, **C**onsent, **C**atheterised bladder; **E**ngaged head (≤1/5, at/below spines); **P**ain relief and **P**elvis adequate; **S**killed operator and **S**upport (neonatal and CS back-up).

Stopping rules "3-2-1-20": **3** pulls, **2** pop-offs, **1** pull without descent, **20** minutes (maximum).

Scalp swellings "Caput Crosses and is there at Birth; CepHalhaematoma Halts at sutures and comes in Hours; SubGaleal Gallops across and is Grave."

OSCE tips

> **EXAM FAVOURITE**
> In the ventouse OSCE, examiners watch for three things above all: **(1) checking fifths abdominally**, **(2) correct cup placement over the flexion point with a finger sweep for trapped tissue**, and **(3) traction with contractions along the pelvic curve, with clearly stated stopping rules**. Say the rules out loud.

- Say "I would call for help and a newborn resuscitator" at the start; it is a critical step.
- Empty the bladder and state it.
- After delivery, do not forget oxytocin 10 IU IM, the PR exam, vitamin K and the neonatal observation plan.

12. References

1. Royal College of Obstetricians and Gynaecologists. Assisted Vaginal Birth (Green-top Guideline No. 26). 2020. https://www.rcog.org.uk/guidance/browse-all-guidance/green-top-guidelines/assisted-vaginal-birth-green-top-guideline-no-26/
2. Assisted Vaginal Birth: Green-top Guideline No. 26 (BJOG abstract). BJOG; 2020. https://pubmed.ncbi.nlm.nih.gov/32346983/
3. Vacuum-assisted vaginal delivery (review). PubMed. https://pubmed.ncbi.nlm.nih.gov/19399290/
4. RANZCOG. Subgaleal Haemorrhage (clinical guidance statement). https://ranzcog.edu.au/wp-content/uploads/Subgaleal-Haemorrhage.pdf
5. An engineering perspective of vacuum assisted delivery devices in obstetrics: a review. Proc Inst Mech Eng H; 2021. https://pmc.ncbi.nlm.nih.gov/articles/PMC7780266/
6. Intrapartum Ultrasound in Vacuum Operative Delivery: A Comprehensive Review and Proposal of the Novel Ultrasound Flexion Point Method. PubMed Central. https://pmc.ncbi.nlm.nih.gov/articles/PMC13024969/
7. Prospective assessment of vacuum deliveries from midpelvic station in a tertiary care university hospital. PubMed Central. https://pmc.ncbi.nlm.nih.gov/articles/PMC8594828/
8. Yim et al. Compliance with clinical guidelines increases the safety of vacuum-assisted delivery. Aust N Z J Obstet Gynaecol; 2023. https://obgyn.onlinelibrary.wiley.com/doi/10.1111/ajo.13545
9. Instrumental Vaginal Deliveries. Obgyn Key. https://obgynkey.com/instrumental-vaginal-deliveries/
10. Merck Manual Professional Edition. Birth Injuries (cephalohematoma, subgaleal haemorrhage). https://www.merckmanuals.com/professional/pediatrics/perinatal-problems/birth-injuries
11. Ministry of Health Uganda. Essential Maternal and Newborn Clinical Care Guidelines for Uganda. 2022. https://library.health.go.ug/sexual-and-reproductive-health/essential-maternal-and-newborn-clinical-care-guidelines-uganda-may
12. Ministry of Health Uganda. Uganda Clinical Guidelines 2023. https://library.health.go.ug/sites/default/files/resources/Uganda%20Clinical%20Guidelines%202023.pdf
13. World Health Organization. Managing Complications in Pregnancy and Childbirth: A Guide for Midwives and Doctors (vacuum extraction section). 2nd edition. Geneva: WHO; 2017.
14. World Health Organization. WHO recommendations: intrapartum care for a positive childbirth experience. Geneva: WHO; 2018.
15. Knight M et al. Prophylactic antibiotics in the prevention of infection after operative vaginal delivery (ANODE): a multicentre randomised controlled trial. The Lancet; 2019.
16. O'Mahony F, Hofmeyr GJ, Menon V. Instruments for assisted vaginal delivery. Cochrane Database of Systematic Reviews; 2010.
17. Glynn M, Drake WM, editors. Hutchison's Clinical Methods: An Integrated Approach to Clinical Practice. 24th edition. Edinburgh: Elsevier; 2018.

SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Where is the flexion point?	On the **sagittal suture about 3 cm in front of the posterior fontanelle** (about 6 cm behind the anterior fontanelle). Centring the cup here gives a **flexing median application**, so the **suboccipitobregmatic (9.5 cm)** diameter presents.
2	List the prerequisites for ventouse delivery.	**Fully dilated** cervix; **membranes ruptured**; **cephalic**, **position known**; head **≤1/5 palpable** with bony station **at or below the spines**; no 3+ moulding; **empty bladder**; **consent**; **analgesia**; trained operator; **neonatal resuscitator**; **back-up for CS**.
3	Working vacuum pressure for ventouse.	About **0.8 kg/cm²** = about **500–600 mmHg** = about **60–80 kPa** (the green zone on a Kiwi gauge). Start at about 0.2 kg/cm² to check for trapped tissue, then increase.
4	When should a ventouse be abandoned?	• **No descent** with a correctly applied pull • Delivery **not imminent after 3 pulls** • Cup **detaches twice** • About **15–20 min** total application • Fetal deterioration with birth not imminent Then go to **CS**. Avoid sequential instruments.
5	RCOG classification of assisted vaginal birth.	**Outlet**: scalp visible without separating the labia; ≤45° rotation. **Low**: skull at **+2 cm or below**, not on the pelvic floor. **Mid**: head **1/5 palpable**, station **0 to +1**. **High** (≥2/5): not appropriate. Do a CS.
6	Absolute contraindications to ventouse.	Cervix not fully dilated; head **≥2/5 palpable** or above the spines; **unknown position**; **face, brow or breech**; **gestation <32 weeks**; suspected **CPD**; **fetal bleeding disorder** or bone demineralisation; no back-up or no trained operator.
7	Direction and technique of traction.	Pull **only with contractions and maternal pushing**, **perpendicular to the cup**, along the **pelvic curve** (down and back, then horizontal, then up as the head crowns). Non-pulling hand: **thumb on the cup, finger on the scalp** to feel descent and slippage. **No rocking or twisting.**
8	Why is a finger swept around the cup after initial vacuum?	To make sure **no cervix or vaginal wall is trapped** under the cup. Trapped tissue causes **cervical and vaginal lacerations**, haemorrhage and cup failure.
9	Soft vs rigid (metal) vs Kiwi OmniCup cups.	**Soft (silicone)**: less scalp trauma, **higher failure** (especially OP). Best for OA outlet or low. **Metal (Malmström or Bird)**: more successful, more chignon and cephalhaematoma. The **Bird posterior cup** is for OP. **Kiwi OmniCup**: rigid plastic, disposable, integral hand pump, low profile, reaches the flexion point in **all positions**.
10	Vacuum vs forceps: key differences.	**Vacuum**: fails more often; more **cephalhaematoma** and **retinal haemorrhage**; **less maternal perineal trauma** (OASI); less analgesia; needs maternal effort. **Forceps**: more successful; more **OASI**, vaginal trauma and **facial injury**; usable at 32–36 weeks, for face presentation and the after-coming head.
11	Caput vs cephalhaematoma vs subgaleal haemorrhage.	**Caput**: scalp oedema, **crosses sutures**, present **at birth**, gone in 1–2 days. **Cephalhaematoma**: **subperiosteal**, **limited by sutures**, appears **hours later**, resolves in weeks. **Subgaleal**: **subaponeurotic**, **crosses sutures**, **enlarges**, can cause **hypovolaemic shock**.

#	Question	Answer
12	Signs of subgaleal haemorrhage after ventouse.	A diffuse **boggy, fluctuant, shifting** scalp swelling crossing sutures (forehead to nape, around the ears); **increasing head circumference** (about 1 cm ≈ 40 mL of blood); **pallor, tachycardia, hypotonia, poor feeding, shock**; falling Hb; DIC.
13	Management of subgaleal haemorrhage.	Urgent neonatal team; **hourly** obs and **head circumference**; Hb and coagulation; **NS 10–20 mL/kg** IV bolus; **blood transfusion**; **FFP** for coagulopathy; **vitamin K 1 mg**; NICU transfer. Mortality can be 12–25%.
14	Maternal and fetal indications for assisted vaginal birth.	**Fetal**: suspected fetal compromise in the second stage. **Maternal**: delay in the second stage (primigravida >2 h, multipara >1 h active pushing in Ugandan teaching); **exhaustion**; medical need to avoid prolonged Valsalva (cardiac disease, hypertensive crisis, cerebrovascular disease, proliferative retinopathy, myasthenia).
15	How do you determine the position of the fetal head on VE?	Follow the **sagittal suture** to a fontanelle. **Posterior fontanelle**: small, **Y-shaped (3 sutures)**. **Anterior**: large, **diamond (4 sutures)**. If the posterior fontanelle is anterior, the position is **OA**; if the anterior fontanelle is anterior, it is **OP**. If caput obscures them, feel an **ear** (the pinna folds towards the occiput) or use **ultrasound**.
16	Moulding grades and their significance.	**+1** bones touching; **+2** overlapping but reducible; **+3** overlapping and **irreducible**. +3 moulding with a head still palpable abdominally suggests **CPD**, so **no ventouse; do a CS**.
17	Maternal aftercare after ventouse.	**Oxytocin 10 IU IM** (third stage); inspect and **PR exam** for OASI, then repair; a **single IV dose of amoxicillin-clavulanate** (ANODE) or per the unit protocol; analgesia; **void within about 6 h** (retention risk); VTE assessment; **debrief**.
18	Why is ventouse important in Uganda's EmONC framework?	**Assisted vaginal delivery is one of the 7 BEmONC signal functions**, expected at HC III and above. It is the **least performed** signal function, so women with second-stage delay are referred for risky **second-stage CS**.
19	What does a 'trial in theatre' mean, and when is it used?	An assisted vaginal birth attempted **in theatre with everything ready for an immediate CS** (anaesthesia, consent for both). Used when the failure risk is higher: **mid-cavity**, 1/5 palpable, **OP**, **EFW >4 kg**, BMI >30, prolonged labour.
20	The A–J mnemonic for vacuum extraction.	**A**sk for help / consent / anaesthesia; **B**ladder empty; **C**ervix fully dilated; **D**etermine position and station (think shoulder Dystocia); **E**quipment ready; **F**lexion point; **G**entle traction; **H**alt (stopping rules); **I**ncision (episiotomy?); **J**aw reachable, so remove the cup.
21	Fifths palpable + bony station: which combinations mean outlet, low, mid and high, and where is each done?	**Outlet**: 0/5, scalp visible without parting the labia: delivery room. **Low**: 0/5, bony skull **+2 or below**: delivery room. **Mid**: **1/5**, station **0 to +1**: **trial in theatre** by an experienced operator. **High**: **≥2/5** or above the spines: **no ventouse, CS**. If abdomen and VE disagree, **trust the abdomen**.
22	In what order do you examine before a ventouse, and what must each step show?	1. **Abdomen**: head **0/5 or 1/5**, contractions, bladder, fetal size, FHR after a contraction 2. **Empty the bladder** (in-out catheter) 3. **VE**: cervix **fully dilated**, membranes **ruptured**, **bony station at or below the spines**, **position known** (sutures and fontanelles), moulding **not +3**, caput, asynclitism, pelvis, liquor 4. Check that fifths and station **agree**

#	Question	Answer
23	Primigravida, pushing 2.5 h after a long augmented labour. Head 1/5, station 0/+1, caput ++, moulding +2, direct OP, EFW 3.9 kg. Delivery-room ventouse?	**No.** This is a **mid-cavity OP** head after a long labour: high failure risk. Options are a **trial in theatre** by an experienced operator with a **posterior cup** (Kiwi OmniCup or Bird posterior), consent for CS and the anaesthetist ready, or **CS directly**. If the head were **≥2/5 palpable** or moulding **+3**: **CS only**.
24	What happens in the first 5 minutes when you are called to a second-stage woman with fetal bradycardia?	Check the mother (pulse, BP, bleeding, uterine tenderness: exclude **abruption/rupture**). Listen to the **FHR for a full minute after a contraction**. **Call for help** and a **newborn resuscitator**. Left lateral, **stop oxytocin**, IV fluids, **empty the bladder**. Get the vacuum set **checked in parallel**, and **tell theatre early**. Then examine: if all prerequisites are met and the head is low, **ventouse now**; if not, **Category 1 CS**.
25	Post-ventouse checks for mother and baby.	**Mother**: oxytocin 10 IU IM, uterus and lochia every 15 min for 1 h, EBL, inspect the genital tract, **PR exam** before and after repair, **void within about 6 h**, single-dose antibiotic, analgesia, VTE assessment, debrief. **Baby**: Apgars, **scalp** exam, **head circumference at birth**, **vitamin K 1 mg IM**, observations **at least hourly for the first few hours**, then 2–4-hourly for 12–24 h. A rising head circumference (**1 cm ≈ 40 mL** blood) means **subgaleal haemorrhage**.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 19-year-old primigravida has been pushing for 2.5 hours at an HC IV. Abdominally the head is 2/5 palpable. On VE the cervix is fully dilated, there is a large caput at the introitus, and moulding is +3 (irreducible). The FHR is 170 bpm. The midwife asks if she should apply the ventouse. What is the MOST appropriate management?

- A Apply a metal Malmström cup and pull with the next contraction
- B Apply a Kiwi OmniCup as a trial in the labour room
- C Start an oxytocin infusion to improve pushing
- D Resuscitate (IV fluids, catheter, antibiotics) and proceed to caesarean section
- E Apply fundal pressure with the next contraction

2. During a ventouse delivery for a direct occipito-anterior position, the consultant asks the student where the centre of the cup should be placed. What is the CORRECT answer?

- A On the sagittal suture about 3 cm in front of the posterior fontanelle
- B Directly over the anterior fontanelle
- C Midway between the anterior and posterior fontanelles
- D About 3 cm in front of the anterior fontanelle
- E Over whichever part of the scalp is most accessible at the introitus

3. A ventouse is applied for fetal distress in a primigravida with the head at +2, LOA. After the first pull the head descends a little. On the second pull the cup detaches; it is reapplied and detaches again with no further descent. The FHR is now 90 bpm. What is the MOST appropriate next step?

- A Reapply the cup a third time at higher pressure
- B Change to forceps immediately, performed by the junior doctor
- C Continue pulling between contractions to speed things up
- D Wait 30 minutes for the head to descend with maternal pushing
- E Abandon the ventouse and proceed to Category 1 CS, warning the team of a possibly impacted head

4. A 28-year-old P3 at an HC III has been pushing for 1.5 hours. The head is 0/5 palpable, the cervix fully dilated, station +3, position ROA, moulding +1. The FHR shows late decelerations. The nearest hospital with theatre is 2 hours away. A trained midwife and a working Kiwi OmniCup are available. What is the BEST management?

Advanced

- A Refer immediately to hospital for caesarean section
- B Start oxytocin augmentation and reassess in 1 hour
- C Proceed with ventouse delivery at the HC III after consent, bladder emptying and analgesia, with a newborn resuscitator present
- D Apply fundal pressure to expedite the birth
- E Wait for spontaneous delivery, as she is multiparous

5. Three hours after a difficult ventouse delivery (4 pulls, 2 detachments), a term baby is pale and tachycardic (HR 180) with poor tone. There is a diffuse, boggy swelling from the forehead to the nape, crossing suture lines, and the ears are pushed forward. The head circumference has increased by 2 cm since birth. What is the MOST appropriate management? Advanced

A Reassure the mother: this is a cephalhaematoma and will resolve in weeks

B Urgent neonatal review; IV normal saline 10–20 mL/kg, blood transfusion, coagulation screen and vitamin K, with hourly head circumference and observations

C Reassure: this is caput succedaneum and will resolve in 48 hours

D Aspirate the swelling with a needle to relieve the pressure

E Start phototherapy for expected jaundice

6. A primigravida at Mengo needs an assisted delivery for prolonged second stage (head 0/5, +2, OA). She asks why the doctor prefers the vacuum rather than forceps. Which statement about vacuum compared with forceps is CORRECT?

A Vacuum causes less maternal perineal trauma, including fewer third- and fourth-degree tears

B Vacuum is more likely to achieve vaginal birth than forceps

C Vacuum causes fewer cephalhaematomas than forceps

D Vacuum causes fewer neonatal retinal haemorrhages

E Vacuum can safely be used at 30 weeks' gestation

7. On VE in the second stage, a student feels a small Y-shaped fontanelle near the sacrum and a large diamond-shaped fontanelle behind the symphysis. The head is 0/5 palpable at +2. The consultant decides on a ventouse. Which cup and application is MOST appropriate? Advanced

A A soft silicone cup placed just inside the introitus

B An anterior metal Malmström cup placed over the diamond-shaped fontanelle

C A soft cup placed over the sagittal suture midway between the fontanelles

D A posterior cup (e.g. Kiwi OmniCup or Bird posterior cup) slid back to the flexion point near the posterior fontanelle

E Any cup, pulling upward from the first contraction

8. Six hours after a ventouse delivery with a second-degree tear, a 24-year-old has not passed urine. She has suprapubic discomfort and a palpable bladder. What is the MOST appropriate management?

A Wait until 24 hours postpartum before intervening

B Give furosemide 40 mg IV

C Restrict oral fluids

D Give a stat dose of antibiotics for a UTI

E Catheterise, measure the residual volume, and leave an indwelling catheter for 24–48 h if the volume is large

9. A student is assessing a multiparous woman before a possible ventouse. Which ONE finding is a contraindication to proceeding?

A Membranes ruptured with clear liquor

B Position left occipito-anterior

C Cervix 9 cm with a thick anterior lip

D Moulding +1

E Head 0/5 palpable with the bony skull at +2

10. The cup is correctly applied at the flexion point and the working pressure has been reached. How should traction be performed?

A Continuous traction without stopping until the head delivers

B With each contraction and maternal push, perpendicular to the cup, along the pelvic curve, with the other hand's thumb on the cup and a finger on the scalp

C Rock the cup from side to side to loosen the head

D Twist the cup to rotate the occiput anteriorly

E Pull steeply upward from the first pull to lift the head over the perineum

11. After an uncomplicated ventouse delivery with an intact perineum, the registrar asks what antibiotic prophylaxis, if any, is recommended (RCOG, following the ANODE trial). What is the BEST answer? Advanced

A No antibiotics are ever needed after assisted birth

B Oral amoxicillin 500 mg 8-hourly for 7 days

C Ceftriaxone 1 g IV daily for 5 days

D Metronidazole 400 mg orally 8-hourly for 5 days

E A single IV dose of amoxicillin-clavulanate as soon as possible after birth (or the unit's single-dose alternative)

12. A trainee places the ventouse cup with its leading edge on the anterior fontanelle and begins traction. What is the MOST likely consequence?

A Faster delivery because the head is lifted more directly

B Increased flexion of the head

C No effect, as cup position does not matter once vacuum is achieved

D Deflexion of the head, so that a larger diameter presents, with a higher risk of failure, pop-offs and scalp trauma

E Guaranteed autorotation to occipito-anterior

13. A 3.4 kg baby was delivered by uncomplicated ventouse (2 pulls, no detachments) and has good Apgars. Only a chignon is visible. What is the MOST appropriate neonatal plan?

A Vitamin K 1 mg IM, and regular observations of HR, RR, colour, tone, feeding and scalp swelling or head circumference (frequently in the first hours, then over 12–24 h), teaching the mother what to watch for

B Discharge mother and baby 1 hour after birth without further observation

C Routine skull X-ray to exclude fracture

D Routine cranial ultrasound before discharge

E Aspirate the chignon to reduce swelling

14. After a ventouse delivery with a right mediolateral episiotomy, the student notices a tear extending towards the anus. What is the MOST important step before starting the repair?

A Start suturing the skin immediately to reduce bleeding

B Give antibiotics and leave the tear to heal by secondary intention

C Examine carefully with good light and analgesia, including a rectal examination, to assess the anal sphincter and grade the tear

D Pack the vagina and discharge her the next day

E Apply a pressure dressing and reassess in 24 hours

15. A 30-year-old P1 with rheumatic mitral stenosis (NYHA class III) is fully dilated with the head 0/5 palpable, station +3, OA, and good contractions. The obstetrician wants to avoid prolonged maternal pushing. What is the MOST appropriate plan? Advanced

A Encourage vigorous pushing for up to 2 hours
B Shorten the second stage with an elective low-cavity assisted vaginal birth, with minimal maternal effort
C Immediate caesarean section at full dilatation
D Apply fundal pressure to avoid maternal pushing
E Start an oxytocin infusion to strengthen contractions

Answers & explanations

1. D · Resuscitate (IV fluids, catheter, antibiotics) and proceed to caesarean section

The head is **2/5 palpable** (not engaged) despite a caput at the introitus, with **+3 moulding** after a prolonged second stage. That means **obstructed labour/CPD**, so ventouse is **contraindicated**. The management is resuscitation and **CS** (**D**). **A** and **B** would fail and risk subgaleal haemorrhage and maternal trauma; the caput misleads about station. **C**: oxytocin in obstruction risks uterine rupture. **E**: fundal pressure is harmful and does not overcome CPD.

2. A · On the sagittal suture about 3 cm in front of the posterior fontanelle

The **flexion point** lies on the sagittal suture about **3 cm anterior to the posterior fontanelle**. Centring the cup there flexes the head and presents the smallest diameter (**A**). **B**, **C** and **D** are too anterior (**deflexing** application), so a larger diameter presents, with more failure and pop-offs. **E** is a classic error: the most accessible part is often too anterior.

3. E · Abandon the ventouse and proceed to Category 1 CS, warning the team of a possibly impacted head

Two detachments and **no descent with a pull**, plus fetal bradycardia, mean **abandon now** and deliver by **Category 1 CS**. Warn the team to prepare for disimpaction (**E**). **A**: repeated reapplication and higher pressure increase the risk of subgaleal haemorrhage and ICH. **B**: **sequential instruments** increase neonatal trauma, and an inexperienced operator makes it worse. **C**: traction without contractions is ineffective and traumatic. **D**: the fetus is bradycardic, so waiting is dangerous.

4. C · Proceed with ventouse delivery at the HC III after consent, bladder emptying and analgesia, with a newborn resuscitator present

All prerequisites are met (0/5, +3, known OA position, fully dilated) and there is a fetal indication. **Assisted vaginal delivery is a BEmONC function** of HC III. A skilled ventouse now (**C**) is far safer than a 2-hour referral with a distressed fetus. **A** delays birth and exposes her to a second-stage CS. **B**: augmenting a multipara with fetal distress is dangerous. **D**: fundal pressure is not recommended (uterine rupture, fetal injury). **E** ignores fetal compromise and prolonged second stage.

5. B · Urgent neonatal review; IV normal saline 10–20 mL/kg, blood transfusion, coagulation screen and vitamin K, with hourly head circumference and observations

A swelling that **crosses sutures and is enlarging**, with **shock** and a rising head circumference (2 cm ≈ 80 mL of blood), is **subgaleal haemorrhage**. It is life-threatening and needs **volume replacement, transfusion, correction of coagulopathy and NICU care** (**B**). **A**: a cephalhaematoma does not cross sutures and doesn't cause shock. **C**: caput is present at birth and does not enlarge. **D**: aspiration introduces infection and does not stop bleeding. **E** ignores the immediate threat.

6. A · Vacuum causes less maternal perineal trauma, including fewer third- and fourth-degree tears

Cochrane evidence shows vacuum causes **less maternal perineal trauma** (fewer OASI and vaginal lacerations) and needs less analgesia (**A**). **B** is wrong: vacuum is **more likely to fail**. **C** and **D** are wrong: vacuum causes **more** cephalhaematoma and retinal haemorrhage. **E**: vacuum is **contraindicated below 32 weeks**.

7. D · A posterior cup (e.g. Kiwi OmniCup or Bird posterior cup) slid back to the flexion point near the posterior fontanelle

The posterior fontanelle is posterior, so this is **occipito-posterior (OP)**. The **flexion point lies far back** in the posterior vagina. A low-profile **posterior cup** is needed to reach it (**D**). **A** and **C**: a soft cup is prone to failure in OP, and both placements are too anterior. **B**: placing it over the anterior fontanelle is a deflexing application (a larger diameter). **E**: pulling upward early extends the head and drags against the symphysis.

8. E · Catheterise, measure the residual volume, and leave an indwelling catheter for 24–48 h if the volume is large

Postpartum urinary retention is common after assisted birth (perineal pain, swelling, trauma). An overdistended bladder can cause long-term **detrusor damage**. Catheterise, measure, and leave an indwelling catheter if the residual is large (**E**). **A** risks overdistension injury. **B**: a diuretic worsens distension. **C** does not treat retention. **D**: there is no evidence of infection.

9. C · Cervix 9 cm with a thick anterior lip

The cervix must be **fully dilated**. A **9 cm cervix with an anterior lip** means she is not in the second stage, and applying a cup risks **cervical laceration** and failure (**C**). **A**, **B**, **D** and **E** are all favourable prerequisites: ruptured membranes, a known OA position, minimal moulding, and a low engaged head.

10. B · With each contraction and maternal push, perpendicular to the cup, along the pelvic curve, with the other hand's thumb on the cup and a finger on the scalp

Traction is **intermittent, synchronised with contractions and pushing**, **perpendicular to the cup**, and follows the **pelvic curve** (down, horizontal, then up). The **finger-thumb** position detects descent and slippage (**B**). **A**: continuous traction exhausts the scalp and increases trauma. **C** and **D**: rocking and twisting break the seal and cause scalp lacerations and subgaleal bleeding; the head should **autorotate**. **E**: early upward traction extends the head and pulls against the symphysis.

11. E · A single IV dose of amoxicillin-clavulanate as soon as possible after birth (or the unit's single-dose alternative)

The **ANODE trial** showed that a **single IV dose of amoxicillin-clavulanate** after assisted vaginal birth significantly reduces maternal infection. The RCOG recommends it (**E**). Where it is unavailable, Ugandan units use a single-dose alternative per protocol. **A** is outdated since ANODE. **B**, **C** and **D** are prolonged courses with no evidence of benefit, and they drive resistance.

12. D · Deflexion of the head, so that a larger diameter presents, with a higher risk of failure, pop-offs and scalp trauma

A cup placed too far anteriorly is a **deflexing application**. Traction extends the head, so the **occipitofrontal (11.5 cm)** diameter presents instead of the suboccipitobregmatic (9.5 cm). The result is **failure, detachments and scalp trauma** (**D**). **A**, **B**, **C** and **E** are wrong because correct flexion-point placement is the most important determinant of success.

13. A · Vitamin K 1 mg IM, and regular observations of HR, RR, colour, tone, feeding and scalp swelling or head circumference (frequently in the first hours, then over 12–24 h), teaching the mother what to watch for

Every ventouse baby needs **structured observation** for **subgaleal haemorrhage**, which may develop over hours, plus routine **vitamin K** (**A**). **B** risks missing a life-threatening bleed. **C** and **D** are not routine and are only indicated if there are clinical concerns. **E**: a chignon is benign and resolves in hours, and aspiration causes infection.

14. C · Examine carefully with good light and analgesia, including a rectal examination, to assess the anal sphincter and grade the tear

Instrumental delivery and episiotomy raise the risk of **obstetric anal sphincter injury (OASI)**. A systematic examination with a **PR exam** is essential to **grade the tear** and ensure that third- and fourth-degree tears are repaired by an experienced operator in theatre (**C**). **A** risks missing and inadequately repairing OASI, which leads to faecal incontinence and fistula. **B**, **D** and **E** leave a significant tear unrepaired.

15. B · Shorten the second stage with an elective low-cavity assisted vaginal birth, with minimal maternal effort

Significant cardiac disease is a **maternal indication** for assisted vaginal birth, to **avoid the haemodynamic stress of prolonged Valsalva** (**B**). Forceps need the least maternal effort; a ventouse with limited pushing is an alternative when forceps skills are unavailable. **A** is dangerous for a woman with mitral stenosis. **C**: CS at full dilatation with the head at +3 carries higher surgical and haemodynamic risk than a low assisted birth. **D**: fundal pressure is harmful. **E**: contractions are already good, and oxytocin adds fluid and haemodynamic effects.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Prolonged second stage with fetal distress at an HC IV

A 22-year-old primigravida at 39 weeks is admitted to an HC IV in Luweero. She has been fully dilated for 2 hours and 15 minutes, pushing for most of that time. She is exhausted. The FHR drops to 100 bpm with each contraction and recovers slowly. Abdominally, contractions are 4 in 10 minutes, strong. You are the medical officer.

1. What will you look for on examination to decide whether a ventouse is possible?

- **Abdomen**: **fifths palpable** (need 0/5 or 1/5), contractions, a **full bladder**, fetal size.
- **VE**: cervix **fully dilated** (no lip), **membranes ruptured**, **station of the bony skull** (not the caput), **position** (sutures and fontanelles, ear), **caput and moulding** grade, liquor colour, pelvis.
- Maternal pulse, BP and temperature; hydration.

2. Findings: head 0/5, station +2, LOA, caput +, moulding +1, clear liquor. Are the prerequisites met? What else must you do before applying the cup?

Yes. It is a **low-cavity, non-rotational (≤45°)** assisted birth with a fetal and a maternal indication. Before applying: **explain and obtain consent**; call a **midwife assistant** and an **HBB-trained newborn resuscitator** (bag and mask ready); **empty the bladder** (in-out catheter); **analgesia** (perineal infiltration with 10 mL lidocaine 1%, or a pudendal block); **check the equipment** (pump, tubing, cup, vacuum test); lithotomy position; asepsis; oxytocin 10 IU drawn up; confirm theatre back-up is possible if it fails.

3. Describe the application and traction.

Insert the cup and slide it back to the **flexion point** (on the sagittal suture about 3 cm in front of the posterior fontanelle). Create low vacuum (about 0.2 kg/cm²) and **sweep a finger around the cup** to exclude trapped cervix or vagina. Increase to **about 0.8 kg/cm² (about 600 mmHg)**. With the next contraction and her push, pull **perpendicular to the cup along the pelvic curve** (downward first), with the **thumb on the cup and a finger on the scalp**. Stop between contractions and check the FHR. Expect descent with each pull. Let the head autorotate; never twist.

4. The head descends with each pull and crowns on the third pull. What next?

Guard the perineum and consider a **selective mediolateral episiotomy** only if the perineum is very tight and tearing is imminent. **Release the vacuum when the jaw is reachable**, then deliver the head, shoulders and body (anticipate shoulder dystocia). **Oxytocin 10 IU IM**, controlled cord traction, and uterine massage. Inspect for tears and do a **PR exam**, then repair. Note the **number of pulls (3), detachments (0) and time**.

5. Outline the post-delivery plan for mother and baby.

Mother: obs and uterine tone; watch for PPH; a single dose of IV prophylactic antibiotic per protocol (co-amoxiclav if available); paracetamol and diclofenac; **void within 6 h** with the volume checked; VTE risk assessment and early mobilisation; debrief and documentation. **Baby**: Apgars; **scalp check** (chignon); **head circumference**; **vitamin K 1 mg IM**; **regular observations** (HR, RR, colour, tone, feeding, scalp swelling) for 12–24 h; early breastfeeding; teach the mother the danger signs (spreading swelling, pallor, poor feeding).

Take-away: When the prerequisites are met (0/5, low station, known OA position, fully dilated), a skilled ventouse at an HC IV delivers a distressed baby in minutes and avoids a second-stage CS. Correct flexion-point placement and traction with contractions are the keys.

Case 2: The ventouse that should become a caesarean

At a regional referral hospital, a 31-year-old primigravida (BMI 34, EFW 3.9 kg) has been fully dilated for 3 hours after oxytocin augmentation. The head is 1/5 palpable. On VE there is a large caput at +2, and the sutures are difficult to feel. The intern proposes a ventouse in the labour room.

1. What concerns you about this situation?

> - **1/5 palpable** means **mid-cavity**: the skull may be higher than the caput suggests.
> - **Large caput, sutures difficult to feel**: the **position is uncertain** (possibly OP or deflexed).
> - **Prolonged labour with augmentation**, **BMI 34**, **EFW 3.9 kg**: a higher risk of **failure, shoulder dystocia and PPH**.
>
> These are all factors for a **trial in theatre**, not a labour-room ventouse.

2. How would you establish the position, and what is a 'trial in theatre'?

> Feel for an **ear** (the pinna folds towards the occiput) or, better, use **transabdominal ultrasound** (fetal orbits face anteriorly in OP; the spine is posterior) or a transperineal scan. A **trial in theatre** means moving to theatre with **spinal anaesthesia**, **consent for both assisted delivery and CS**, and the theatre team scrubbed, so that you can move to **CS immediately** if it fails. A senior operator should conduct it.

3. Ultrasound shows a direct OP position. In theatre the senior applies a Kiwi OmniCup at the flexion point. After 2 pulls there is minimal descent and one detachment. What now?

> **Abandon.** There is no clear progressive descent after 2 pulls plus a detachment, delivery is not imminent, and there is a big baby and a prolonged labour. **Do not reapply repeatedly, and do not switch to forceps** (sequential instruments raise the risk of neonatal ICH and trauma). **Proceed to CS immediately** under the same spinal.

4. What difficulties do you anticipate at the CS and how will you manage them?

> - **Deeply impacted head**: an assistant **disimpacts vaginally** (push method, cupped hand, pushing the head up and flexing it), or use the **reverse breech (pull) method**. Consider tocolysis (terbutaline 0.25 mg SC) to relax the uterus.
> - **Extensions** of the uterine incision into the broad ligament and **PPH**: good exposure, Green-Armytage clamps, oxytocin, TXA 1 g IV.
> - **Bladder injury** (high, oedematous bladder).
> - **Neonatal** trauma and asphyxia: the neonatal team is present; observe for subgaleal haemorrhage (vacuum traction was applied).

5. What should be documented and discussed afterwards?

> Document the **prerequisite findings** (fifths, station, position by ultrasound, caput and moulding), the reason for the trial in theatre, the **cup type, number of pulls, detachments and times**, the decision to abandon, and the CS findings. **Debrief** the woman (why it happened, what it means for the next birth: a trial of labour is possible). Review at the **perinatal audit** if there were adverse outcomes.

Take-away: Mid-cavity station, uncertain position, a big caput, prolonged labour and a large baby predict failure. Clarify the position (with ultrasound if possible), do the trial in theatre, stop early if there is no descent, and never go on to sequential instruments.

Case 3: The pale baby on the postnatal ward

You are called to the postnatal ward at Mengo at 06:00. A term baby girl (3.1 kg) was born at 01:30 by ventouse after 5 pulls with 2 cup detachments. The mother says the baby 'is not breastfeeding and her head is getting bigger'. The baby is pale, with a HR of 184 bpm, RR 60, capillary refill 4 seconds, and she is floppy.

1. What features on examination of the head will help you make the diagnosis?

> Look for a **diffuse, boggy, fluctuant swelling** that **crosses suture lines**, extending from the **orbits to the nape** and **around the ears** (ears pushed forward). It **shifts with head position** and may pit. **Measure the head circumference** and compare it with the birth measurement: each 1 cm rise ≈ 40 mL of blood. Compare with a **cephalhaematoma** (confined by sutures, not shocked) and **caput** (present at birth, decreasing).

2. What is the diagnosis and why is it so dangerous?

> **Subgaleal (subaponeurotic) haemorrhage.** Bleeding from ruptured emissary veins into the large potential space between the **galea aponeurotica and the periosteum** can hold **several hundred mL**, i.e. most of a newborn's blood volume (about 80–90 mL/kg, so about 270 mL here). She is showing **hypovolaemic shock** (tachycardia, pallor, prolonged capillary refill, hypotonia). **Consumptive coagulopathy** may follow. Mortality is significant.

3. Describe your immediate management.

> 1. **Call the paediatrician or neonatal team**; move to the special care unit.
> 2. **ABC**: oxygen, warmth, IV or intraosseous access.
> 3. Bloods: **Hb/PCV, grouping and cross-match**, coagulation screen, glucose.
> 4. **Normal saline 10–20 mL/kg IV bolus**, then **packed red cells or fresh whole blood** (10–20 mL/kg, repeated as needed; O-negative in an emergency).
> 5. **FFP** for coagulopathy; **vitamin K 1 mg IV/IM** (check it was given).
> 6. **Hourly head circumference**, HR, BP (if possible), cap refill and urine output.
> 7. Avoid head bandaging (it doesn't tamponade and hides the swelling). Refer to the NICU if unstable.
> 8. Later, watch for jaundice (phototherapy) and seizures (ICH).

4. What went wrong in the delivery, and how could this have been prevented?

> - The ventouse **continued beyond the stopping rules**: **5 pulls and 2 detachments** (the rules are 3 pulls, 2 detachments, no descent means stop).
> - Possible **incorrect cup placement** (deflexing or paramedian), rocking or twisting, or pulling without contractions.
> - **Neonatal observation was not structured**: 4.5 hours passed before recognition.
>
> Prevention: meet the prerequisites, apply the cup at the flexion point, **respect the abandon criteria**, avoid sequential instruments, and give **every ventouse baby regular observations** of head circumference and vital signs, with the mother taught what to report.

5. How should this be followed up at a system level?

> **Open disclosure** to the parents. Incident report and **perinatal death/near-miss review (MPDSR)**. Retrain staff on ventouse technique and the stopping rules (simulation). Introduce a **ventouse documentation proforma** and a **post-vacuum neonatal observation chart** on the postnatal ward.

Take-away: Subgaleal haemorrhage is a silent killer after ventouse. Respect the stopping rules (3 pulls, 2 pop-offs, no descent means stop), observe every vacuum baby with regular head measurements, and treat shock with volume and blood urgently.

References

1. Assisted Vaginal Birth (Green-top Guideline No. 26). *Royal College of Obstetricians and Gynaecologists*, 2020. https://www.rcog.org.uk/guidance/browse-all-guidance/green-top-guidelines/assisted-vaginal-birth-green-top-guideline-no-26/
2. Assisted Vaginal Birth: Green-top Guideline No. 26 (BJOG abstract). *BJOG / PubMed*, 2020. https://pubmed.ncbi.nlm.nih.gov/32346983/
3. Vacuum-assisted vaginal delivery (review). *PubMed*. https://pubmed.ncbi.nlm.nih.gov/19399290/
4. Subgaleal Haemorrhage (clinical guidance statement). *RANZCOG*. https://ranzcog.edu.au/wp-content/uploads/Subgaleal-Haemorrhage.pdf
5. An engineering perspective of vacuum assisted delivery devices in obstetrics: a review. *Proceedings of the Institution of Mechanical Engineers, Part H (via PMC)*, 2021. https://pmc.ncbi.nlm.nih.gov/articles/PMC7780266/
6. Intrapartum Ultrasound in Vacuum Operative Delivery: A Comprehensive Review and Proposal of the Novel Ultrasound Flexion Point Method. *PubMed Central (NCBI)*. https://pmc.ncbi.nlm.nih.gov/articles/PMC13024969/
7. Prospective assessment of vacuum deliveries from midpelvic station in a tertiary care university hospital. *PubMed Central (NCBI)*. https://pmc.ncbi.nlm.nih.gov/articles/PMC8594828/
8. Compliance with clinical guidelines increases the safety of vacuum-assisted delivery (Yim et al.). *Australian and New Zealand Journal of Obstetrics and Gynaecology*, 2023. https://obgyn.onlinelibrary.wiley.com/doi/10.1111/ajo.13545
9. Instrumental Vaginal Deliveries. *Obgyn Key*. https://obgynkey.com/instrumental-vaginal-deliveries/
10. Birth Injuries (cephalohematoma, subgaleal haemorrhage). *Merck Manual Professional Edition*. https://www.merckmanuals.com/professional/pediatrics/perinatal-problems/birth-injuries
11. Essential Maternal and Newborn Clinical Care Guidelines for Uganda. *Ministry of Health Uganda*, 2022. https://library.health.go.ug/sexual-and-reproductive-health/essential-maternal-and-newborn-clinical-care-guidelines-uganda-may
12. Uganda Clinical Guidelines 2023. *Ministry of Health Uganda*, 2023. https://library.health.go.ug/sites/default/files/resources/Uganda%20Clinical%20Guidelines%202023.pdf
13. Managing Complications in Pregnancy and Childbirth: A Guide for Midwives and Doctors (vacuum extraction section). *World Health Organization*, 2017.
14. Prophylactic antibiotics in the prevention of infection after operative vaginal delivery (ANODE): a multicentre randomised controlled trial (Knight et al.). *The Lancet*, 2019.
15. Instruments for assisted vaginal delivery (O'Mahony, Hofmeyr, Menon). *Cochrane Database of Systematic Reviews*, 2010.

A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.